Machine Learning

The Exponential Data Explosion

- Nowadays, the formation of data around the world is increasing exponentially from almost every organization such as educational institutions, e-commerce industries, social networking sites, healthcare sector and many other organizations.
- In every **3 - 4** years, the volume of data is doubling.

Year	Global Data Volume
2023	120 ZB
2024	147 ZB
2025	181 ZB
2026 (forecast)	221 ZB

So there is a need of a powerful tool that extracts information from data generated from various sectors.

- Data mining is an emerging interdisciplinary tool that is looking for obtaining useful hidden patterns amongst the datasets.
- Steps involved in data mining process utilizes the ability of Statistics, Artificial Intelligence, Machine Learning and Database technology.

Transforming Data into Information

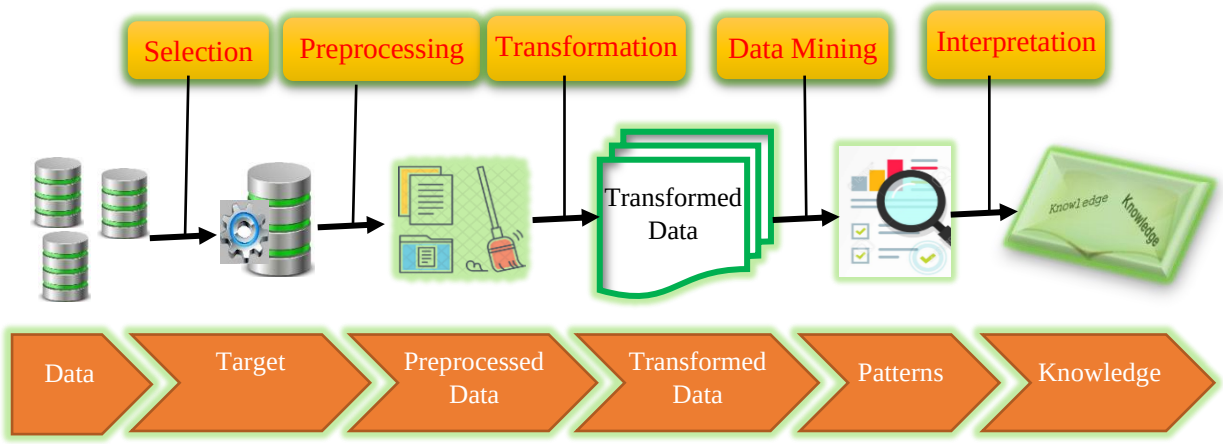


Figure : Steps involve in Data mining process

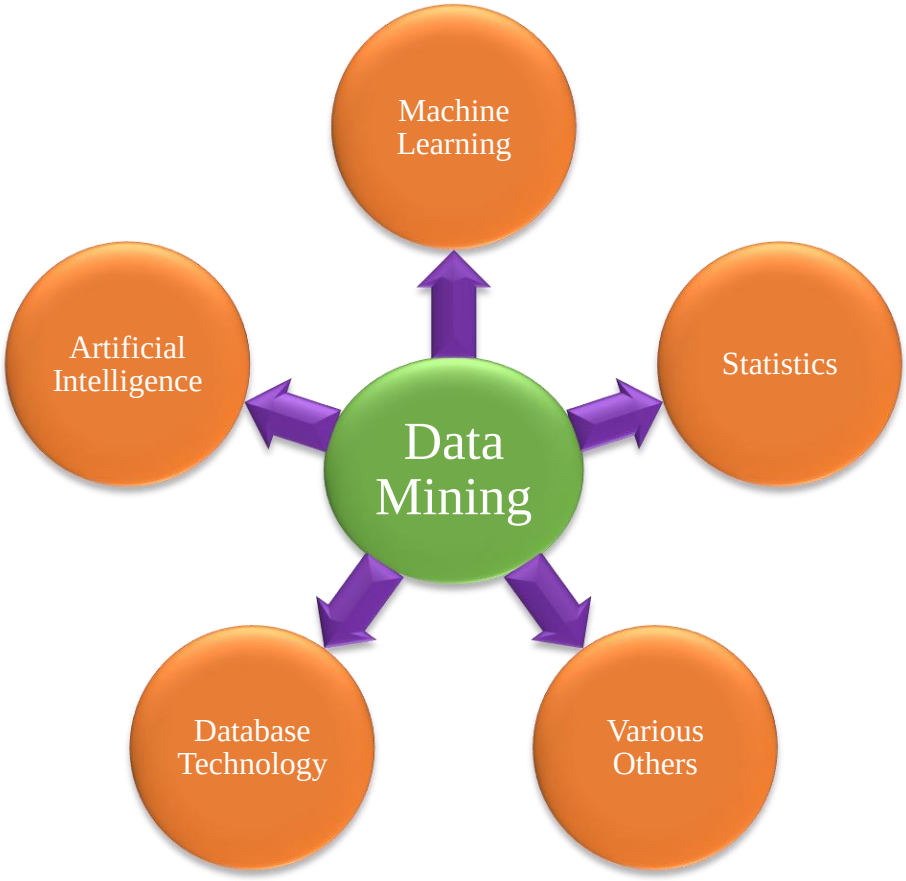


Figure: Data mining utilizes methods from several areas

Machine Learning

- ✓ Machine learning (ML) is a technique of data analysis that provides a system which can automatically learn and improve from past experiences without being explicitly programmed for it.

- ✓ Instead of manually writing rules for every possible situation, ML algorithms automatically discover relationships in data and improve their performance through experience.

Example

Suppose we want to detect spam emails.

Traditional Programming

Rules are manually written:

- If the email contains "Lottery" → Spam
- If it contains "Congratulations" → Spam
- If the sender is “unknown” → Spam

The programmer must keep updating the rules.

Machine Learning

Thousands of emails labeled as "**Spam**" and "**Not Spam**" are provided to the algorithm.

The model automatically learns:

- Which words are associated with spam
- Which combinations of words indicate spam
- How likely a new email is to be spam

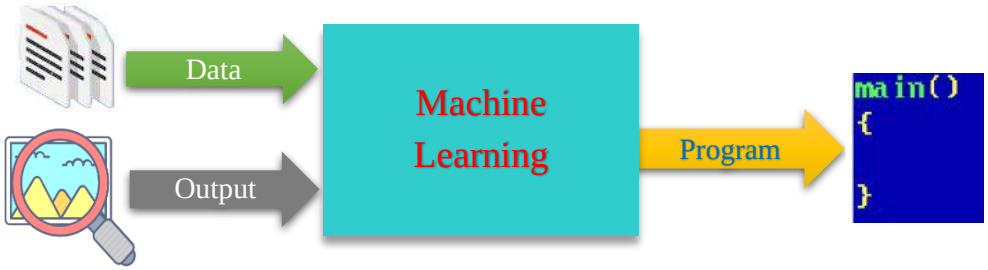
This makes ML more flexible and scalable.

Conventional v/s ML Programming

In ***traditional programming***, we provided ***input*** as **data** and **program** (logic) to the computer for producing the desired output.



Whereas in machine learning, we feed ***input*** as **data** as well as **output** to the computer to create the program.



Characteristics of Machine Learning

Machine Learning systems have several important characteristics:

- ✓ **Learning from Data**
  Instead of following fixed rules, ML systems learn patterns directly from data.

- ✓ **Generalization**
  A good model performs will not only on training data but also on unseen data.
- ✓ **Adaptability**
  Models can be retrained with new data to improve performance.
- ✓ **Pattern Recognition**
  ML identifies hidden relationships in data that humans may overlook.
- ✓ **Prediction Capability**
  After training, models can predict future outcomes.

Types of Machine Learning

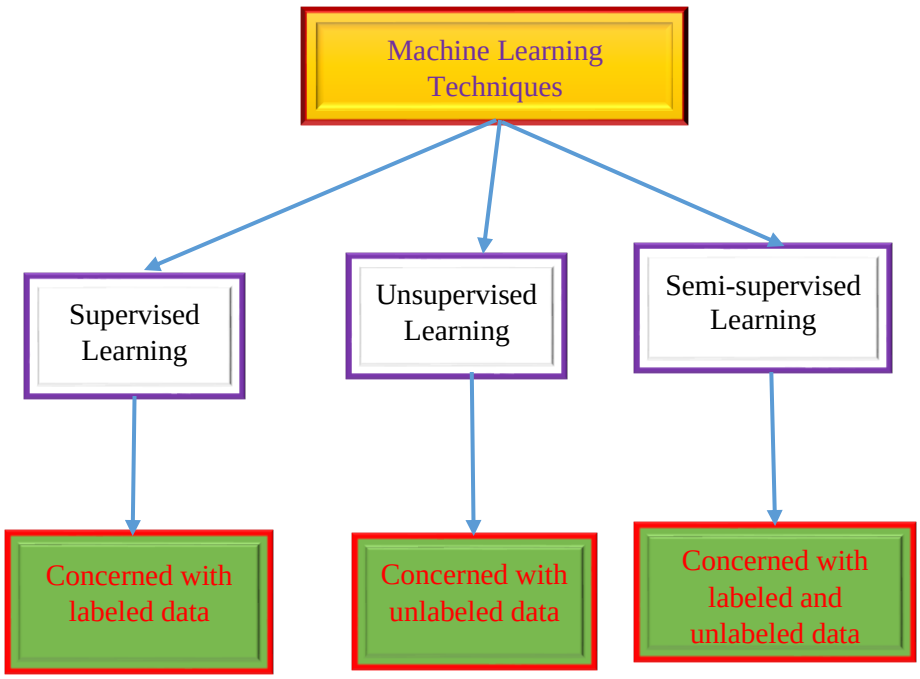


Figure: Taxonomy of Machine Learning Techniques

Taxonomy of Machine Learning Techniques

Machine learning techniques are mainly classified into three categories such as Supervised, Unsupervised and Semi-supervised Machine Learning.

Supervised Machine Learning

Supervised learning uses labeled data, where both **input** and **correct output** are known.

Example:

SL. NO	REVIEW	SENTIMENT
1	GREAT PHONE	Positive
2	BAD BATTERY	Negative

Applications:

- Spam Detection
- Sentiment Analysis
- Disease Prediction
- House Price Prediction

- In supervised machine learning, a **model** is **built** that maps a set of **input data** to the corresponding **output data**.
- The model is **trained** using ***training data*** that has a predetermined ***class label*** (outputs). During training, the parameter value of models is appropriately adjusted to obtain a proper level of accuracy.
- Finally, the **model** can ***predict*** the **label** for ***new input*** (test data) after adequate training.

Classification is the example of supervised machine learning.

Classification:

Basic terminology used in classification-

Dataset: A collection of instances is called a dataset that contains essential features.

Features: A feature is an essential property of the object that is fed to the ML algorithm to learn the model.

Class: It indicates the data belongs to a particular category.

Model: A model is a mathematical function of a real-world problem that performs a specific operation. It requires input (training) data to learn the model.

Validation: Validation is the process which is used to verify the accuracy of the trained model prediction.

Common Algorithms:

- Decision Tree
- Random Forest
- Naïve Bayes
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Artificial Neural Network (ANN)

Dataset

Day	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No
15	Sunny	Cool	High	Strong	No
16	Overcast	Hot	High	Strong	Yes
17	Sunny	Hot	Normal	Strong	Yes
18	Sunny	Hot	High	Strong	Yes
19	Rain	Cool	High	Strong	No

Training Data

	X_train				Y_train
Day	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

	Test Data				
	X_test				Y_test
Day	Outlook	Temp	Humidity	Wind	Play
15	Sunny	Cool	High	Strong	No
16	Overcast	Hot	High	Strong	Yes
17	Sunny	Hot	Normal	Strong	Yes
18	Sunny	Hot	High	Strong	Yes
19	Rain	Cool	High	Strong	No

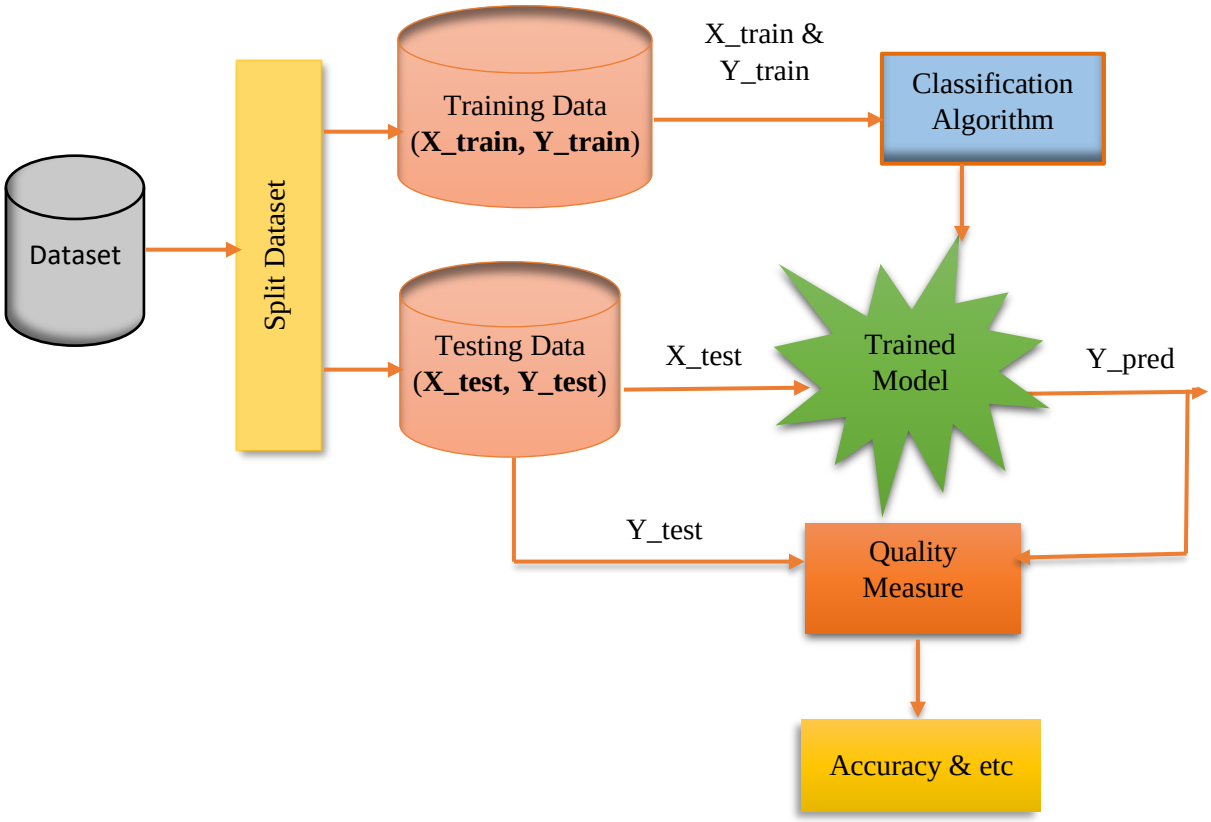


Figure: Working of classification method

- Classification uses the mathematical function to develop a model that can learn how to classify the unknown data/objects into a predefined set of classes.
- During the model development stage, the classification algorithm determines the association between the predictors (X_train) and their target values (Y_train) of training data.
- These associations are summed up in a model, which would then be able to apply to a different test instance (X_test) in which the class labels are missing.
- The model is evaluated by comparing the predicted values (Y_pred) to target values (Y_test) of test data.

Example

Unsupervised Machine Learning

- Unsupervised learning works with unlabeled data.
- It learns a few properties from the data and during the prediction of the class of new data, trying to recognize similarities which split the data into a specific category.
- **Clustering**, **Association Rule Mining** and **Feature Selection** are examples of unsupervised machine learning.

Clustering:

Goal:

- Discover hidden structures
- Group similar data

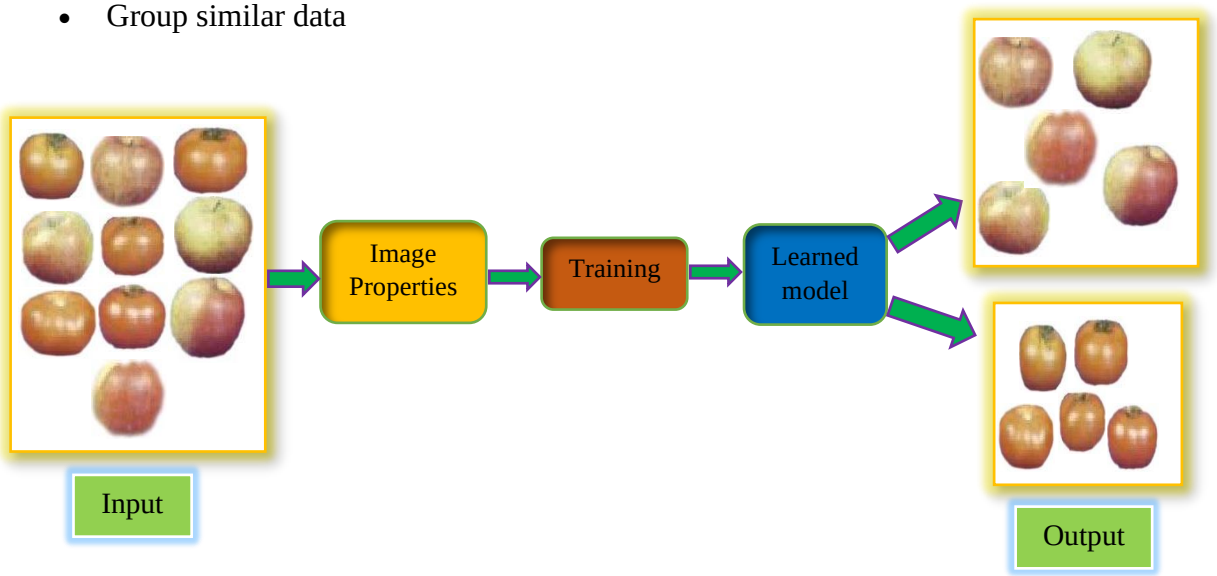


Figure: Process of unsupervised learning

Algorithms:

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN

Applications:

- Customer Segmentation
- Market Analysis
- Anomaly Detection

Clustering Numerical Example

Semi-supervised machine learning

- Semi-supervised learning (SSL) methods have become popular in the last few years. It takes input as a mixture of both labelled and unlabeled data.
- The goal of SSL is to categorize the unlabeled data using the labelled data.
- For example, every text documents are not labelled, so in text document classification, we may use some labelled text document to classify the unlabeled documents.

Real-World Applications of ML

Computer Vision	Facial Recognition, Self-Driving Cars, Quality Inspection
Finance	Fraud Detection, Credit Scoring, Algorithmic Trading, Risk Assessment
Healthcare	Disease Prediction, Medical Imaging (Cancer Detection), Drug Discovery, Personalized Treatment
Natural Language	Chatbots (ChatGPT), Machine Translation, Sentiment Analysis, Summarization
Agriculture	Crop Yield Prediction, Disease Detection, Precision Farming, Soil Analysis, Weather Forecasting and Smart Irrigation.

Recommendation	Netflix/Amazon Recommendations, Personalized Ads,

Thank You